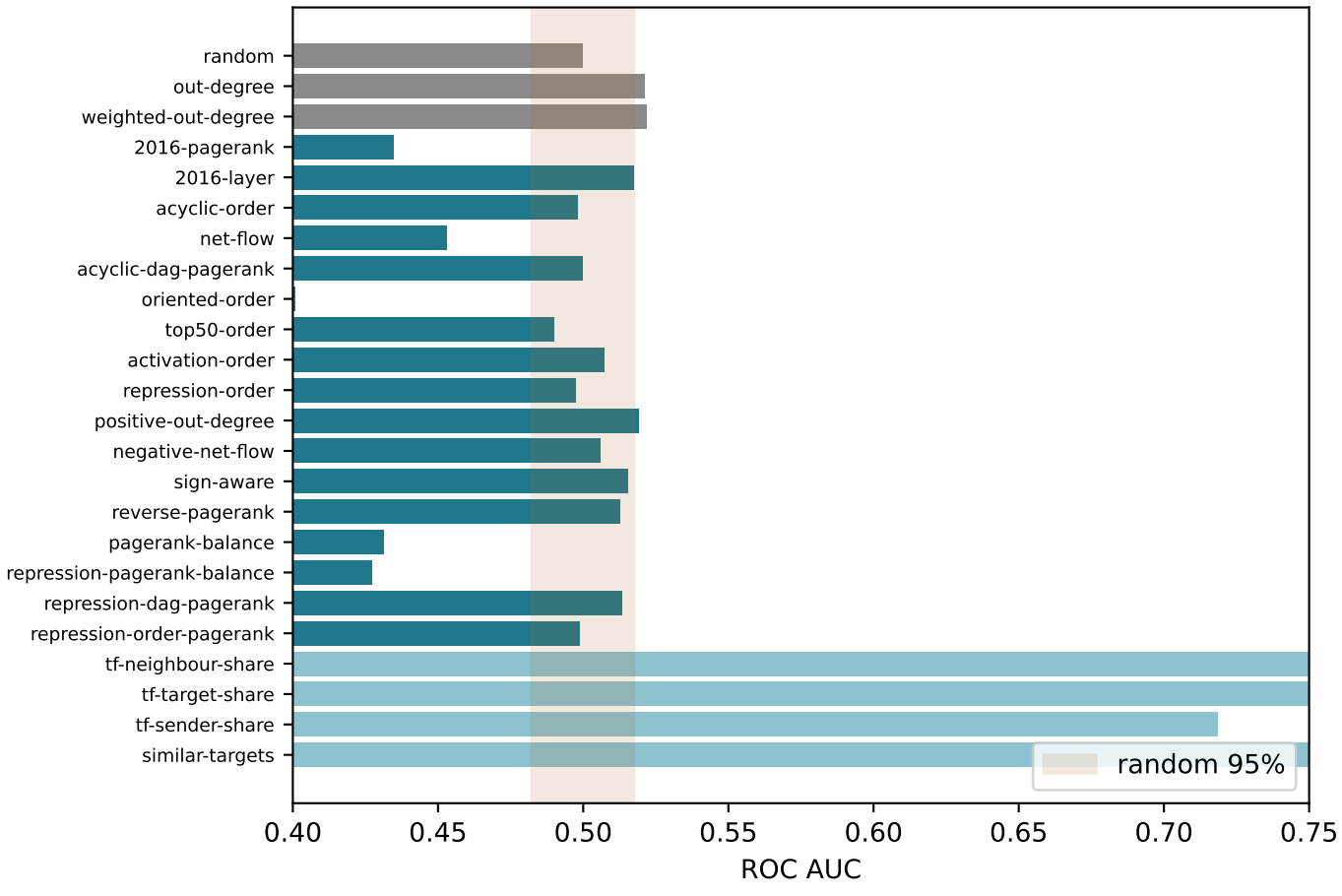
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,832 genes ranked, 1,139 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.6 (4–16)	0.500 (0.482–0.517)				
out-degree	8	0.521	0.748	0.009	-0.001 (-0.002, +0.001)	0.480
weighted-out-degree	7	0.522	0.856	0.007	+0.000 (+0.000, +0.000)	
2016-pagerank	13	0.435	0.156	1.000	-0.087 (-0.109, -0.065)	< 0.001
2016-layer	11	0.517	0.367	0.026	-0.004 (-0.022, +0.012)	0.622
acyclic-order	7	0.498	0.856	0.588	-0.024 (-0.034, -0.015)	< 0.001
net-flow	7	0.453	0.856	1.000	-0.069 (-0.081, -0.057)	< 0.001
acyclic-dag-pagerank	7	0.500	0.856	0.503	-0.022 (-0.028, -0.016)	< 0.001
oriented-order	7	0.401	0.856	1.000	-0.121 (-0.138, -0.104)	< 0.001
top50-order	4	0.490	0.987	0.870	-0.032 (-0.052, -0.012)	< 0.001
activation-order	8	0.507	0.748	0.199	-0.014 (-0.030, +0.000)	0.058
repression-order	11	0.497	0.367	0.624	-0.024 (-0.047, -0.002)	0.024
positive-out-degree †	9	0.519	0.637	0.017	-0.003 (-0.011, +0.006)	0.572
negative-net-flow †	7	0.506	0.856	0.238	-0.016 (-0.031, -0.000)	0.044
sign-aware †	9	0.516	0.637	0.042	-0.006 (-0.014, +0.003)	0.198
reverse-pagerank †	8	0.513	0.748	0.077	-0.009 (-0.014, -0.004)	< 0.001
pagerank-balance †	9	0.431	0.637	1.000	-0.090 (-0.108, -0.073)	< 0.001
repression-pagerank-balance †	9	0.427	0.637	1.000	-0.094 (-0.114, -0.077)	< 0.001
repression-dag-pagerank †	6	0.513	0.935	0.069	-0.008 (-0.028, +0.011)	0.388
repression-order-pagerank †	10	0.499	0.510	0.547	-0.023 (-0.045, -0.001)	0.042
tf-neighbour-share ‡	86	0.751	< 0.001	< 0.001	+0.229 (+0.209, +0.250)	< 0.001
tf-target-share ‡	69	0.754	< 0.001	< 0.001	+0.232 (+0.210, +0.255)	< 0.001
tf-sender-share ‡	84	0.718	< 0.001	< 0.001	+0.197 (+0.175, +0.218)	< 0.001
similar-targets ‡	94	0.754	< 0.001	< 0.001	+0.232 (+0.209, +0.255)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	30,999	0.048 (2.68×)	3.39×	1.96×	0.61 (0.60)
resolve_reciprocals → drop_sinks → prune_triangles	30,969	0.017 (2.01×)	3.31×	1.97×	0.20 (0.19)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	15,208	0.029 (1.35×)	2.79×	1.35×	0.15 (0.15)
absolute	30,999	0.048 (2.68×)			
absolute → minimum_feedback_order	30,845	0.035 (2.32×)			
orient_reciprocals ratio=2.0 → absolute	30,969	0.030 (2.21×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,912	0.026 (2.04×)			
top_outgoing k=50 → absolute	30,999	0.014 (4.17×)			
top_outgoing k=50 → absolute → minimum_feedback_order	30,791	0.009 (3.95×)			
keep_sign sign=1 → absolute	30,925	0.031 (3.38×)			
keep_sign sign=1 → absolute → minimum_feedback_order	30,767	0.021 (3.15×)			
keep_sign sign=-1 → absolute	20,718	0.026 (1.68×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	17,771	0.029 (1.63×)			
keep_sign sign=-1	20,718	0.026 (1.68×)		1.68×	0.16 (0.16)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,512,607 edges (1,268,310 negative), largest strong component 11,832.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,512,607 edges (1,268,310 negative), largest strong component 11,832.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 1,226,030 edges (1,187,520 negative), largest strong component 11,671.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 1,226,030 edges (1,187,520 negative), largest strong component 11,671; DAG keeps 1,083,259 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,512,607 edges (0 negative), largest strong component 11,832; DAG keeps 2,033,065 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,512,607 edges (0 negative), largest strong component 11,832.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,512,607 edges (0 negative), largest strong component 11,832; DAG keeps 2,033,065 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,933,374 edges (0 negative), largest strong component 11,784; DAG keeps 1,792,228 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 463,461 edges (0 negative), largest strong component 11,665; DAG keeps 321,231 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,244,297 edges (0 negative), largest strong component 11,555; DAG keeps 867,202 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,268,310 edges (0 negative), largest strong component 6,024; DAG keeps 1,241,519 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,512,607 edges (1,268,310 negative), largest strong component 11,832.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,512,607 edges (1,268,310 negative), largest strong component 11,832.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,512,607 edges (1,268,310 negative), largest strong component 11,832.

reverse-pagerank: Highest reverse PageRank (raw graph)